# WITH J UNIVERSALLY CLOSED THE ETALE NEIGHBOURHOOD CONCLUSION CAN BE SHRUNK SO THAT R' IS EXACTLY THE RESTRICTION OF THE BASE CHANGED GROUPOID

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*Allowed standard prerequisites (editorial).*

- The quasi-finite-over-base etale neighbourhood lemma with conclusions (a) through (g), assuming its conditions (1) through (6) (more-groupoids chapter, Lemma lemma-quasi-finite-over-base, canonical tag 03FL; collected on this lane as L22508). cited at statement lines 2568-2570 and 2577-2578 and proof lines 2586-2594. Produces the etale neighbourhood, the decompositions and u' with (a) through (g), the substrate on which the shrink to (h) is performed; its conclusions (b), (c), (d), (e) are used individually in the proof.
- Criterion for a subgroupoid decomposition to be the restriction of a groupoid relation to an open (groupoids chapter, Lemma groupoids-lemma-restrict-groupoid-relation, canonical tag 02VD). cited at proof lines 2619-2621. Identifies Rest = empty set with the condition that R' is the restriction of S' times_S R to the open U', giving conclusion (h).

*Source reference.* Original lemma 03FL: sources/stacks-chow/tags.txt, source lines 4496–4496. Cited at statement lines 2568-2570 for assumptions (1)-(6) and at proof lines 2586-2594 for the etale neighbourhood, decompositions and conclusions (a)-(g) that the proof builds on.

*Source reference.* Original lemma 02VD: sources/stacks-chow/tags.txt, source lines 3764–3764. Cited at proof lines 2619-2621: Rest = empty set is exactly the condition that R' is the restriction of S' times_S R to U', giving conclusion (h).

**Lemma 0.2.** *Let $S$ be a scheme. Let $(U, R, s, t, c)$ be a groupoid scheme over $S$. Let $p \in S$ be a point, and let $u \in U$ be a point lying over $p$. Assume assumptions (1) – (6) of Lemma 03FL hold as well as*

(7) *$j : R \to U \times_S U$ is universally closed*[1].

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[1] In view of the other conditions this is equivalent to requiring $j$ to be proper.

*Then we can choose* $(S', p') \to (S, p)$ *and decompositions* $S' \times_S U = U' \amalg W$ *and* $S' \times_S R = R' \amalg W'$ *and* $u' \in U'$ *such that (a) – (g) of Lemma 03FL hold as well as*

(h) $R'$ *is the restriction of* $S' \times_S R$ *to* $U'$.

*Proof.* We apply Lemma 03FL for the groupoid $(U, R, s, t, c)$ over the scheme $S$ with points $p$ and $u$. Hence we get an étale neighbourhood $(S', p') \to (S, p)$ and disjoint union decompositions

$$S' \times_S U = U' \amalg W, \quad S' \times_S R = R' \amalg W'$$

and $u' \in U'$ satisfying conclusions (a), (b), (c), (d), (e), (f), and (g). We may shrink $S'$ to a smaller neighbourhood of $p'$ without affecting the conclusions (a) – (g). We will show that for a suitable shrinking conclusion (h) holds as well. Let us denote $j'$ the base change of $j$ to $S'$. By conclusion (e) it is clear that

$$j'^{-1}(U' \times_{S'} U') = R' \amalg Rest$$

for some open and closed $Rest$ piece. Since $U' \to S'$ is finite by conclusion (d) we see that $U' \times_{S'} U'$ is finite over $S'$. Since $j$ is universally closed, also $j'$ is universally closed, and hence $j'|_{Rest}$ is universally closed too. By conclusions (b) and (c) we see that the fibre of

$$(U' \times_{S'} U' \to S') \circ j'|_{Rest} : Rest \longrightarrow S'$$

over $p'$ is empty. Hence, since $Rest \to S'$ is closed as a composition of closed morphisms, after replacing $S'$ by $S' \setminus \mathrm{Im}(Rest \to S')$, we may assume that $Rest = \emptyset$. And this is exactly the condition that $R'$ is the restriction of $S' \times_S R$ to the open subscheme $U' \subset S' \times_S U$, see Groupoids, Lemma 02VD and its proof. □